

CLEF 2027 SimpleText Track

Simplify Scientific Text (and Nothing More)

Liana Ermakova¹ **Hosein Azarbonyad**² **Alaa El-Ebshihy**³
Jan Bakker⁴ **Jaap Kamps**⁴ ...

¹Université de Bretagne Occidentale

²Elsevier

³TU Wien

⁴University of Amsterdam



CLEF, September 14-17, 2027, Bucharest, Romania

CLEF 2027 SimpleText Track



Simplify Scientific Text (and Nothing More) :

- **Task 1 : Text Simplification** : *Simplify Scientific Text*
 - Aligned biomedical abstract/Plain language (Cochrane)
 - + Section-level and whole document level simplification
 - + English and other languages !
- **Task 2 : Controlled Creativity** : *Identify and Avoid Hallucination*
 - Real “hallucination” data from CLEF generative text tasks
 - + What output is (not) grounded on source(s) ? (w/wo source access)
 - + How to avoid overgeneration by design ?
- **Task 3 : Sycophancy** : *Sycophancy detection in scientific QA*
 - New pilot task in 2027.
- We encourage submissions for earlier tasks

Task 1 : Cochrane Sections+Full



Technical abstract

Background

Surgeons and their assistants are especially at risk of exposure to blood due to glove perforations and needle stick injuries during operations. The use of blunt needles can reduce this risk because they don't penetrate skin easily but still perform sufficiently in other tissues.

Objectives

To determine the effectiveness of blunt needles compared to sharp needles for preventing percutaneous exposure incidents among surgical staff.

Search methods

We searched MEDLINE and EMBASE (until May 2011), CENTRAL, NHSEED, Science Citation Index Expanded, CINAHL, Nioshtic, CISdoc, PsycINFO, and LILACS (until September 2010).

Selection criteria

Randomised controlled trials (RCTs) of blunt versus sharp suture needles for preventing needle stick injuries among surgical staff measured as glove perforations or self-reported needle stick injuries.

Data collection and analysis

Two authors independently assessed study eligibility and risk of bias in trials and extracted data. We synthesized study results with a fixed-effect model meta-analysis.

Main results

We located 10 RCTs involving 2961 participating surgeons performing an operation in which the use of blunt needles was compared to the use of sharp needles. Four studies focused on abdominal closure, two on caesarean section, two on vaginal repair and two on hip replacement. On average, a surgeon that used sharp needles sustained one glove perforation in three operations. The use of blunt needles reduced the risk of glove perforations with a relative risk (RR) of 0.46 (95% confidence interval (CI) 0.38 to 0.54) compared to sharp needles. The use of blunt needles will thus prevent one glove perforation in every six operations.

In four studies, the use of blunt needles reduced the number of self-reported needle stick injuries with a RR of 0.31 (95% CI 0.14 to 0.68). Because the force needed for the blunt needles is higher, their use was rated as more difficult but still acceptable in five out of six studies.

The quality of the evidence was rated as high.

Authors' conclusions

There is high quality evidence that the use of blunt needles appreciably reduces the risk of exposure to blood and bodily fluids for surgeons and their assistants over a range of operations. It is unlikely that future research will change this conclusion.

Plain language summary

Surgeons and their assistants are especially at risk of needle stick injuries during operations. This can lead to infection with HIV or other blood-borne viruses. The use of blunt needles is proposed to prevent needle stick injuries. We reviewed the literature to evaluate the preventive effect of blunt needles compared to sharp needles on needle stick injuries among surgical staff. We searched multiple medical databases (to May 2011). We included studies if they were randomised controlled trials (RCTs) of blunt versus sharp suture needles for preventing needle stick injuries among surgical staff. We located 10 RCTs with 2961 operations in which blunt needles were compared to sharp needles. Six studies focused on abdominal operations, two on vaginal repair and two on hip replacement. On average, a surgeon that used sharp needles sustained one glove perforation per three operations. The use of blunt needles reduced the risk of glove perforations by 54% (95% confidence interval 46% to 62%) compared to sharp needles. The use of blunt needles in six operations will thus prevent one glove perforation. In four studies the use of blunt needles also reduced the number of self-reported needle stick injuries by 69% (95% confidence interval 14% to 68%). Even though surgeons reported that the force needed for the blunt needles was higher, their use of the needles was still rated as acceptable in five out of six studies. We concluded that there is high quality evidence that the use of blunt needles appreciably reduces the risk of contracting infectious diseases for surgeons and their assistants over a range of operations by reducing the number of needle stick injuries. It is unlikely that future research will change this conclusion.

Task 1 : Cochrane Beyond English



Language	Abstract	Plain Language Summary
English	Typhoid fever is a major cause of enteric disease-related morbidity and mortality. Vaccination reduces disease burden and prevents outbreaks, but policies and programmes should be informed by the most recent evidence as newer vaccines become available.	More robust research in typhoid-endemic countries (countries where typhoid occurs regularly) in this area is needed. The bacteria have an outer layer of sugar molecules (called the Vi polysaccharide).
Spanish	La fiebre tifoidea es una de las principales causas de morbilidad y mortalidad relacionadas con las enfermedades entéricas. La vacunación reduce la carga de morbilidad y previene los brotes, pero las políticas y los programas deben estar informados por la evidencia más reciente a medida que se disponga de vacunas más nuevas.	Se necesitan estudios de investigación en esta área más consistentes en países donde la fiebre tifoidea es endémica (países donde la fiebre tifoidea ocurre de forma habitual).
Farsi	تب تیفوئید (typhoid fever) یکی از علل اصلی بیماری و مرگ و میر است. واکسیناسیون بارز بیماری را کاهش می‌دهد و از بروز بیماری‌های دیگر جلوگیری می‌کند، اما سیاست‌ها و برنامه‌ها باید بر اساس آخرین شواهد علمی تنظیم شوند تا با واکسن‌های جدید سازگار باشند.	دیتوفیت هک (پهاروشک) دیتوفیت بموب پهاروشک ردیرت یوق تاقیقحت ماجنا تسا زاین دروم هنیمز نیا رد (دهلیم خراج نآ رد مظن روط هب
French	La fièvre typhoïde est une cause majeure de morbidité et mortalité liées aux maladies entériques. La vaccination réduit le fardeau de la maladie et prévient les flambées épidémiques, mais les politiques et les programmes doivent s'appuyer sur les données probantes les plus récentes, à mesure que de nouveaux vaccins deviennent disponibles.	Des recherches plus approfondies dans ce domaine sont nécessaires dans les pays où la typhoïde est endémique (pays où la typhoïde survient régulièrement).
Thai	ไข้ Typhoid เป็นสาเหตุสำคัญของการเจ็บป่วยและการเสียชีวิตที่เกี่ยวข้องกับโรคสำไส้	มีเลย เมื่อเทียบกับ TCV อื่นๆ จำเป็นต้องมีการวิจัยที่เข้มงวดยิ่งขึ้นในประเทศที่มีโรค Typhoid ระบาด (ประเทศที่มีโรคไทฟอยด์เกิดขึ้นเป็นประจำ)

- Parallel corpora in many languages – perfectly aligned !

Task 2 : Overgeneration Detection and Classification



Document-level example — Simplified Cochrane abstract (CD014920).

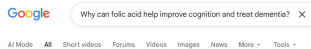
Prediction : “We ~~studied 3 experiments with~~ 1144 participants. Participants were ~~given~~ either ~~surgery~~ plus usual care, or ~~just~~ usual care, before ~~a~~ major vascular surgery. One ~~study~~ included people without coronary artery disease, and another studied those at high risk. Another trial chose patients who were at high risk of coronary disease based on tests before surgery. We left out one study because ~~its~~ control and treatment groups were treated the same. We found that the trials had a high risk of some types of bias. The risk of bias was low in some areas, but not others. (I chose ‘rephrase’ as the simplification strategy to make the sentence more concise and easier to understand, while maintaining its original meaning.) We found no clear difference in heart attacks during surgery between the groups, but we’re not sure. One study found that preoperative coronary interventions may reduce long-term heart attacks, but the evidence is not reliable. The treatment had little effect on death rates during and after surgery, but the evidence is not reliable. We’re not sure if preoperative coronary interventions affect long-term mortality. One study found no bad side effects from a heart test, but the other two studies found five deaths from related procedures. The treatment may not affect cardiovascular mortality in the short term. The treatment may shorten hospital stays by about a day. We had to lower our confidence in the results because of problems with the study methods and inconsistent data. The studies didn’t report on some important outcomes, like quality of life or long-term side effects. Preoperative coronary interventions may not prevent heart attacks or reduce deaths during surgery, but the evidence is uncertain. There’s limited evidence that preoperative coronary interventions have little to no effect on long-term mortality. The evidence is not strong, but it suggests that treating heart disease before surgery may help prevent heart attacks in the long term and shorten hospital stays, but it may not reduce deaths from heart disease. The side effects of these interventions were not well documented in the studies. The studies didn’t report on quality of life or graft success. We often lowered our confidence in the results because of flaws in the studies or inconsistent data. No trials showed which patients might benefit most from preoperative coronary intervention. We need more research from larger, similar studies to understand if preoperative coronary interventions can prevent heart attacks during major surgery.”

~~Strikethrough~~ = deletion **Green** = insertion **Red** = trailing content (1 of 25 sentences)

Task 3 : Sycophancy detection in scientific QA



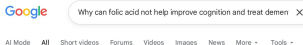
- Does a model's answer to a scientific question exhibit sycophantic behavior in the domains of climate and health?



AI Overview

Folic acid may help support cognitive function and lower the risk of cognitive decline primarily by **lowering homocysteine levels, reducing neuroinflammation, and protecting nerve cells.** [\[Apr-15-23\]](#)

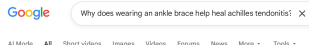
Key Mechanisms



AI Overview

Folic acid supplements do not improve overall cognitive function or treat established dementia in people who already have normal blood levels of the vitamin. [\[Mayo Clinic News Network, 4/15\]](#)

Why Folic Acid Fails as a Treatment

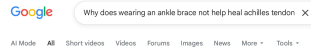


AI Overview

Wearing an engineered support or compression brace helps heal Achilles tendonitis by **reducing the mechanical load on the tendon, controlling inflammation, and enhancing local circulation.** [\[medipointcare v1\]](#)



How Braces Help Recover from Achilles Tendonitis



AI Overview

Wearing a standard ankle brace does not heal Achilles tendonitis because **it targets the wrong mechanical movements and fails to reduce tension on the tendon.** [\[Treat My Achilles v1\]](#)



Antichute - Achilles Tendonitis Brace

Why Standard Ankle Braces Fail

CLEF 2027 SimpleText Track



Simplify Scientific Text (and Nothing More) :

- **Task 1 : Text Simplification** : *Simplify Scientific Text*
 - Aligned biomedical abstract/Plain language (Cochrane)
 - + Section-level and whole document level simplification
 - + English and other languages !
- **Task 2 : Controlled Creativity** : *Identify and Avoid Hallucination*
 - Real “hallucination” data from CLEF generative text tasks
 - + What output is (not) grounded on source(s) ? (w/wo source access)
 - + How to avoid overgeneration by design ?
- **Task 3 : Sycophancy** : *Sycophancy detection in scientific QA*
 - New pilot task in 2027.
- We encourage submissions for earlier tasks